

Chapter Summary

Key Terms

12.1 Graph Basics

- vertex
- edge
- loop
- graph (simple graph)
- multigraph
- adjacent (neighboring)
- degree

12.2 Graph Structures

- complete
- subgraph
- cycle
- cyclic subgraph
- clique

12.3 Comparing Graphs

- isomorphic
- isomorphism
- planar
- nonplanar
- complement
- complementary

12.4 Navigating Graphs

- walk (directed walk)
- trail (directed trail)
- path (directed path)
- closed
- open
- closed walk
- circuit (closed trail)
- directed cycle (closed path)
- coloring (graph coloring)
- n -coloring
- chromatic number

12.5 Euler Circuits

- connected
- component
- disconnected
- Euler circuit
- Eulerian graph
- Chinese postman problem
- Eulerization

12.6 Euler Trails

- algorithm
- Fleury's algorithm
- Euler trail
- bridge
- local bridge

12.7 Hamilton Cycles

- Hamilton cycle, or Hamilton circuit
- n factorial
- weighted graph
- total weight

12.8 Hamilton Paths

- Hamilton path

12.9 Traveling Salesperson Problem

- brute force algorithm
- greedy algorithm
- traveling salesperson problem (TSP)
- brute force method
- nearest neighbor method

12.10 Trees

- acyclic
- tree
- forest
- path graph or linear graph
- star tree
- root
- starlike tree
- caterpillar tree
- lobster tree
- spanning tree
- minimum spanning tree

Key Concepts

12.1 Graph Basics

- Graphs and multigraphs represent objects as vertices and the relationships between the objects as edges.
- The degree of a vertex is the number of edges that meet it and the degree can be zero.
- An edge must have a vertex at each end.
- Multigraphs may contain loops and double edges, but simple graphs may not.

12.2 Graph Structures

- The sum of the degrees of the vertices in a graph is twice the number of edges.
- In a complete graph every pair of vertices is adjacent.
- A subgraph is part of a larger graph.
- Cycles are a sequence of connected vertices that begin and end at the same vertex but never visit any vertex twice.

12.3 Comparing Graphs

- Two graphs are isomorphic if they have the same structure.
- When graphs are relatively small, we can use visual inspection to identify an isomorphism by transforming one graph into another without breaking connections or adding new ones.
- An isomorphism between two graphs preserves adjacency.
- If two graphs differ in number of vertices, number of edges, degrees of vertices, or types of subgraphs, they cannot be isomorphic.
- When the complements of two graphs are isomorphic, so are the graphs themselves.

12.4 Navigating Graphs

- Walks, trails, and paths are ways to navigate through a graph using a sequence of connected vertices and edges.
- Closed walks, circuits, and directed cycles are ways to navigate from a vertex on a graph and return to the same vertex.
- Colorings are a way to organize the vertices of a graph into groups so that no two members of a group are adjacent.

- Maps can be represented with planar graphs, which can always be colored using four colors or fewer.

12.5 Euler Circuits

- A connected graph has only one component.
- The Euler circuit theorem states that an Euler circuit exists in every connected graph in which all vertices have even degree, but not in disconnected graphs or any graph with one or more vertices of odd degree.
- The Chinese postman problem asks how to find the shortest closed trail that visits all edges at least once.
- If an Euler circuit exists, it is always the best solution to the Chinese postman problem.
- Eulerization is the process of adding duplicate edges to a graph so that the new multigraph has an Euler circuit.
- The minimum number of duplicated edges needed to eulerize a graph is half the number of odd vertices or more.

12.6 Euler Trails

- An Euler trail exists whenever a graph has exactly two vertices of odd degree.
- When a bridge is removed from a graph, the number of components increases.
- A bridge is never part of a circuit.
- When a local bridge is removed from a graph, the distance between vertices increases.
- An edge that is part of a triangle is never a local bridge.

12.7 Hamilton Cycles

- A Hamilton cycle is a directed cycle, or circuit, that visits each vertex exactly once.
- Some Hamilton cycles are also Euler circuits, but some are not.
- Hamilton cycles that follow the same undirected cycle in the same direction are considered the same cycle even if they begin at a different vertex.
- The number of unique Hamilton cycles in a complete graph with n vertices is the same as the number of ways to arrange $n - 1$ distinct objects.
- Weighted graphs have a value assigned to each edge, which can represent distance, time, money and other quantities.

12.8 Hamilton Paths

- A Hamilton path visits every vertex exactly once.
- Some Hamilton paths are also Euler trails, but some are not.

12.9 Traveling Salesperson Problem

- A brute force algorithm always finds the ideal solution but can be impractical whereas a greedy algorithm is efficient but usually does not lead to the ideal solution.
- A Hamilton cycle of lowest weight is a solution to the traveling salesperson problem.
- The brute force method finds a Hamilton cycle of lowest weight in a complete graph.
- The nearest neighbor method is a greedy algorithm that finds a Hamilton cycle of relatively low weight in a complete graph.

12.10 Trees

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Videos

12.1 Graph Basics

- [Graph Theory: Create a Graph to Represent Common Boundaries on a Map \(https://openstax.org/r/Graph_Theory\)](https://openstax.org/r/Graph_Theory)

12.2 Graph Structures

- [The Mathematical Secrets of Pascal's Triangle by Wajdi Mohamed Ratemi \(https://openstax.org/r/Wajdi_Mohamed_Ratemi\)](https://openstax.org/r/Wajdi_Mohamed_Ratemi)

12.3 Comparing Graphs

- [Determine If Two Graphs Are Isomorphic and Identify the Isomorphism \(https://openstax.org/r/Determine_If_Two\)](https://openstax.org/r/Determine_If_Two)

12.4 Navigating Graphs

- [Walks, Trails, and Paths in Graph Theory \(https://openstax.org/r/walks_trails_paths\)](https://openstax.org/r/walks_trails_paths)
- [Closed Walks, Closed Trails \(Circuits\), and Closed Paths \(Directed Cycles\) in Graph Theory \(https://openstax.org/r/closed_walks\)](https://openstax.org/r/closed_walks)
- [Coloring Graphs Part 1: Coloring and Identifying Chromatic Number \(https://openstax.org/r/Coloring_and_Identifying\)](https://openstax.org/r/Coloring_and_Identifying)
- [The Four Color Map Theorem – Numberphile \(https://openstax.org/r/The_Four\)](https://openstax.org/r/The_Four)
- [Coloring Graphs Part 2: Coloring Maps and the Four Color Problem \(https://openstax.org/r/Coloring_Maps\)](https://openstax.org/r/Coloring_Maps)
- [Neil deGrasse Tyson Explains the Möbius Strip \(https://openstax.org/r/Neil_de_Grasse\)](https://openstax.org/r/Neil_de_Grasse)

12.5 Euler Circuits

- [Connected and Disconnected Graphs in Graph Theory \(https://openstax.org/r/connected_disconnected_graphs\)](https://openstax.org/r/connected_disconnected_graphs)
- [Recognizing Euler Trails and Euler Circuits \(https://openstax.org/r/Euler_trails_circuits\)](https://openstax.org/r/Euler_trails_circuits)
- [Existence of Euler Circuits in Graph Theory \(https://openstax.org/r/existence_Euler_circuits\)](https://openstax.org/r/existence_Euler_circuits)

12.6 Euler Trails

- [Bridges and Local Bridges in Graph Theory \(https://openstax.org/r/bridges_local_bridges\)](https://openstax.org/r/bridges_local_bridges)
- [Fleury's Algorithm to Find an Euler Circuit \(https://openstax.org/r/Fleury's_algorithm\)](https://openstax.org/r/Fleury's_algorithm)

12.10 Trees

- [The Problem in *Good Will Hunting* by Numberphile \(https://openstax.org/r/Hunting_Numberphile\)](https://openstax.org/r/Hunting_Numberphile)
- [Spanning Trees in Graph Theory \(https://openstax.org/r/Spanning_Trees_in_Graph_Theory\)](https://openstax.org/r/Spanning_Trees_in_Graph_Theory)
- [Use Kruskal's Algorithm to Find Minimum Spanning Trees in Graph Theory \(https://openstax.org/r/Trees_in_Graph_Theory\)](https://openstax.org/r/Trees_in_Graph_Theory)

Formula Review

12.2 Graph Structures

For the Sum of Degrees Theorem, $\text{sum of the degrees} = 2 \times \text{number of edges}$ or $\text{number of edges} = \frac{\text{sum of degrees}}{2}$

The number of edges in a complete graph with n vertices is the sum of the whole numbers from 1 to $n - 1$, $1 + 2 + 3 + \cdots + (n - 1)$.

The number of edges in a complete graph with n vertices is $1 + 2 + 3 + \cdots + (n - 1) = \frac{n(n-1)}{2}$.

12.7 Hamilton Cycles

The number of ways to arrange n distinct objects is $n!$.

The number of distinct Hamilton cycles in a complete graph with n vertices is $(n - 1)!$.

12.9 Traveling Salesperson Problem

- In a complete graph with n vertices, the number of distinct Hamilton cycles is $(n - 1)!$.
- In a complete graph with n vertices, there are at most $\frac{(n-1)!}{2}$ different weights of Hamilton cycles.

12.10 Trees

- The number of edges in a tree graph with n vertices is $n - 1$. A connected graph with n vertices and $n - 1$ edges is a tree graph.

Projects

Everyone Gets a Turn! – Graph Colorings

Let's put your knowledge of graph colorings to work! Your task is to plan a field day following these steps.

1. Select between seven and ten activities for your field day. You can look online for ideas.
2. Create a survey asking for the participants to select the three to five events in which they would most like to participate. Survey between seven and ten people.

3. Use the results of your survey to create a graph in which each vertex represents one of the events. A pair of vertices will be adjacent if there is at least one participant who would like to participate in both events.
4. Find a minimum coloring for the graph. Explain how you found it and how you know the chromatic number of the graph.
5. Use your solution to part d to determine the minimum number of timeslots you must use to ensure that everyone has the opportunity to participate in their top three events.
6. Find the complement of the graph you created. Explain what the edges in this graph represent.

A Beautiful Day in the Neighborhood – Euler Circuits

Let's apply what you have learned to the community in which you live. Using resources such as your county's property appraiser's website, create a detailed graph of your neighborhood in which vertices represent turns and intersections. Represent a large enough part of your community to include no fewer than 10 intersections or turns. Then use your graph to answer the following questions.

1. Label the edges of your graph.
2. Determine if your graph is Eulerian. Explain how you know. If it is not, eulerize it.
3. Find an Euler circuit for your graph. Give the sequence of vertices that you found.
4. What does the Euler circuit you found in part c represent for your community?
5. Describe an application for which this Euler circuit might be used.

Dream Vacation – Hamilton Cycles and Paths

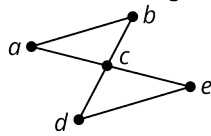
Where in the world would you like to travel most: the Eiffel Tower in Paris, a Broadway musical in New York city, a bike tour of Amsterdam, the Tenerife whale and dolphin cruises in the Canary Islands, the Giza Pyramid in Cairo, or maybe the Jokhang Temple in Tibet? Let's plan your dream vacation!

1. Which four destinations are at the top of your bucket list?
2. Draw a complete weighted graph with five vertices representing the four destinations and your home city, and the weights representing the cost of travel between cities.
3. Use a website (such as [Travelocity \(https://openstax.org/r/travelocity\)](https://openstax.org/r/travelocity)) to find the best airfare between each pair of cities. List the airlines and flight numbers along with the prices. Include cost for ground transportation from the nearest airport if there is no airport at the destination you want to visit.
4. Use the nearest neighbor algorithm to find a Hamilton cycle of low weight beginning and ending in your hometown. What is the weight of this circuit and what does it represent?
5. Use the brute force method to find a Hamilton cycle of lowest weight beginning and ending in your hometown. What is the weight of this circuit? Is it the same or different from the weight of the Hamilton cycle you found in Exercise 4?
6. Suppose that instead of returning home, you planned to move to your favorite location on the list, but you wanted to stop at the other three destinations once along the way. Where would you move? List all Hamilton paths between your hometown and your favorite location.
7. Find the weights of all the Hamilton paths you found in Exercise 6.

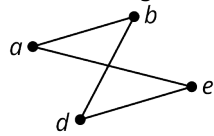
Chapter Review

Graph Basics

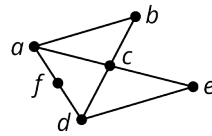
For the following exercises, use the given figure.



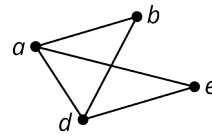
Graph S



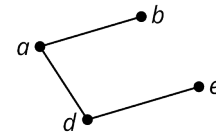
Graph T



Graph U



Graph V

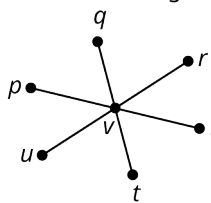


Graph W

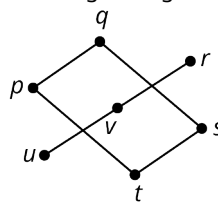
- Determine the number of vertices in Graph S.
- Identify the graph with the fewest edges.
- Name the vertices in Graph U.
- Identify any pairs of vertices in Graph S that are not adjacent.
- Which graphs only has vertices of degree 2?
- Identify the graph in which the sum of the degrees of the vertices is 16.
- Amazon.com has a network of warehouses that are used to move packages around the United States. Delivery trucks from warehouse deliver packages to other locations. These mail trucks also pick up packages to bring back to their home warehouse. Explain how a graph or multigraph might be drawn to model this scenario by identifying the objects that could be represented by vertices and the connections that could be represented by edges. Indicate whether a graph or a multigraph would be a better model.
- Eduardo has two groups of four friends, group A and group B. Within each group, each of the members of the group are friends with each other but not with those in the other group. The individuals in group A have no other friends, but the individuals in group B each has their own group of four friends, and the individuals within those groups are all friends with each other, but with no one outside their own group. Draw a graph to represent the given scenario.

Graph Structures

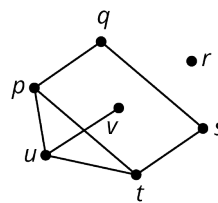
For the following exercises, use the given figure.



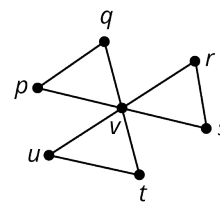
Graph A



Graph B



Graph C

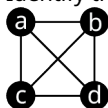


Graph D

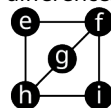
- Explain why Graph B is not a subgraph of Graph C.
- Identify any graphs that have a quadrilateral cyclic subgraph and name the vertices.
- Identify a clique in Graph C by listing its vertices.
- Draw a graph with the following characteristics: largest clique has 4 vertices, a pentagonal cyclic subgraph, exactly two vertices of degree 4.
- How many edges are in a complete graph with 12 vertices?
- How many triangles are in a complete graph with 11 vertices?

Comparing Graphs

- Identify three differences between Graph 1 and Graph 2 that demonstrate the graphs are not isomorphic.

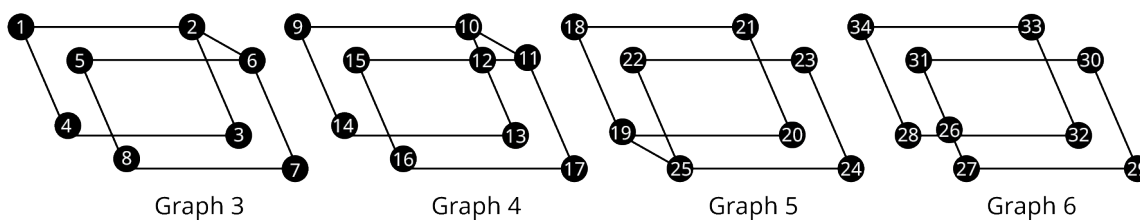


Graph 1



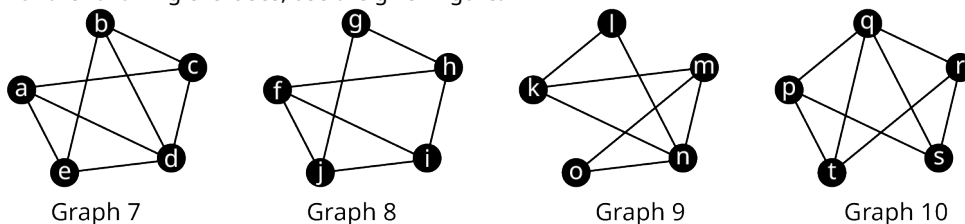
Graph 2

For the following exercises, use the given figure.



16. Determine if Graph 3 is isomorphic to Graph 4. If so, identify a correspondence between the vertices which demonstrates the isomorphism. If not, identify at least two characteristics that verifies this.
17. Determine if Graph 3 is isomorphic to Graph 5. If so, identify a correspondence between the vertices that demonstrates the isomorphism. If not, identify a characteristic that verifies this.
18. Consider Graph 4 and Graph 5. Determine if one graph is a subgraph of the other. If so, give a correspondence between vertices that demonstrates this relationship. If not, identify conflicting characteristics.

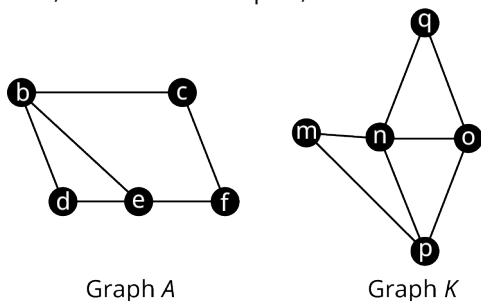
For the following exercises, use the given figure.



19. Draw the complements of Graphs 7 and 9. Determine whether the graphs you drew are isomorphic to each other and explain how you know. Use this information to determine whether Graphs 7 and 8 are isomorphic.
20. Draw the complements of Graphs 7 and 10. Determine whether the graphs you drew are isomorphic to each other and explain how you know. Use this information to determine whether Graphs 7 and 10 are isomorphic.
21. Draw the complements of Graphs 8 and 9. Determine whether the graphs you drew are isomorphic to each other and explain how you know. Use this information to determine whether Graphs 8 and 9 are isomorphic.

Navigating Graphs

For the following exercises, use Graph A in the given figure. Consider each sequence of vertices. Determine if it is only a walk, both a walk and a path, both a walk and a trail, all three, or none of these.



22. $e \rightarrow d \rightarrow b \rightarrow e \rightarrow f$
23. $e \rightarrow b \rightarrow d \rightarrow e \rightarrow f \rightarrow c \rightarrow b \rightarrow d$
24. $e \rightarrow f \rightarrow b \rightarrow d$

For the following exercises, use Graph K in Figure 12.352. Identify each sequence of vertices as a closed walk, circuit (closed trail), and/or directed cycle (closed path). Indicate all that apply.

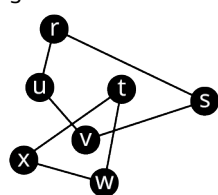
25. $n \rightarrow o \rightarrow q \rightarrow n \rightarrow p \rightarrow m \rightarrow n$
26. $m \rightarrow n \rightarrow o \rightarrow q \rightarrow n \rightarrow p \rightarrow o \rightarrow n \rightarrow m$
27. $p \rightarrow o \rightarrow q \rightarrow n \rightarrow m \rightarrow p$

For the following exercises, use Graphs A and K in Figure 12.352.

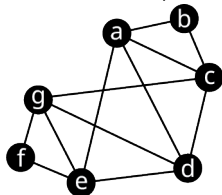
28. Are Graphs A and K planar graphs? What does your answer tell you about the chromatic number of each graph?
29. How many vertices are in the largest complete subgraph of Graph A? What does this tell you about the chromatic number of Graph A?
30. Create a coloring of Graph A, which uses exactly four colors, or explain why it is not possible.
31. Create a coloring of Graph K, which uses exactly two colors, or explain why it is not possible.
32. Determine the chromatic number of Graph A. Give a coloring that supports your conclusion.
33. Determine the chromatic number of Graph K. Give a coloring that supports your conclusion.

Euler Circuits

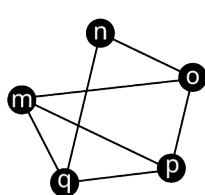
For the following exercises, use the graphs and multigraphs shown. Identify any graphs and/or multigraphs with the given characteristics. If there are none, state so.



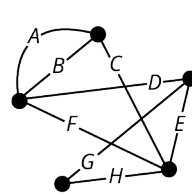
Graph 11



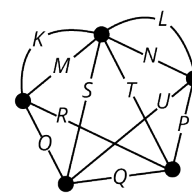
Graph 12



Graph 13



Multigraph 14



Multigraph 15

34. connected
35. disconnected
36. Eulerian

Euler Trails

For the following exercises, use the graphs and multigraphs in Figure 12.354.

37. List the set of vertices for each component in Graph 13.
38. Determine whether the sequence of edges represents an Euler circuit in Multigraph 15:
 $K \rightarrow L \rightarrow N \rightarrow M \rightarrow O \rightarrow S \rightarrow T \rightarrow Q \rightarrow U \rightarrow P \rightarrow R$
39. Find an Euler circuit beginning and ending at vertex g in Graph 12 if one exists. Otherwise, explain how you know such an Euler circuit does not exist.
40. Give an example of a pair of edges that could be duplicated to eulerize Multigraph 14.

For the following exercises, use the graphs and multigraphs in Figure 12.354. Identify any graphs and/or multigraphs with the given characteristics. If there are none, state so.

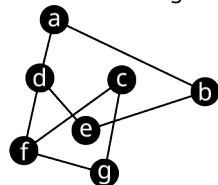
41. Exactly two vertices of odd degree
42. Has an Euler trail
43. Has exactly one local bridge

For the following exercises, use the graphs and multigraphs in Figure 12.354. In each exercise a graph and a sequence of vertices are given.

44. Determine whether the sequence of edges, $A \rightarrow B \rightarrow C \rightarrow H \rightarrow G \rightarrow D \rightarrow F \rightarrow E$, is an Euler trail, an Euler circuit, or neither for the graph. If it is neither, explain why.
45. Suppose that an edge were added to Graph 11 between vertices s and w . Determine if the graph would have an Euler trail or an Euler circuit, and find one.

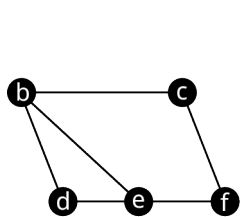
Hamilton Cycles

For the following exercises, refer to the graph in the given figure.

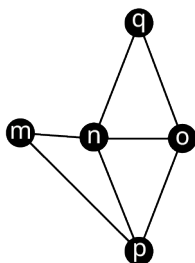


46. A student has been asked to use Fleury's algorithm to construct an Euler trail in the given graph. The student decides to begin the trail at vertex d . Is this a good choice, why or why not?
47. A student who is using Fleury's algorithm to construct an Euler trail has decided to begin with $f \rightarrow d \rightarrow a \rightarrow b \rightarrow \dots$. If the student is off to a good start, help the student by completing the Euler trail. If the student has made an error, explain the error.
48. Use Fleury's algorithm to construct an Euler trail for Graph 16 beginning at the vertex of your choice.

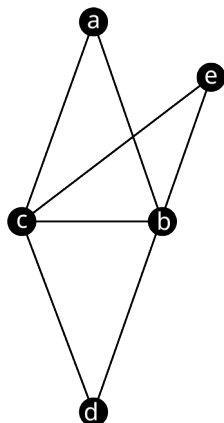
For the following exercises, use the graphs from the given figures to determine whether the sequence of vertices in the given graph is a Hamilton cycle, an Euler circuit, both, or neither.



Graph A



Graph K



Graph F

49. Graph F. $a \rightarrow b \rightarrow e \rightarrow c \rightarrow b \rightarrow d \rightarrow c \rightarrow a$

50. Graph K. $m \rightarrow n \rightarrow q \rightarrow o \rightarrow p \rightarrow m$

51. Graph A. $b \rightarrow d \rightarrow e \rightarrow f \rightarrow c \rightarrow b \rightarrow e$

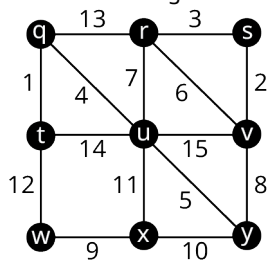
For the following exercises, evaluate the factorial expression for the given value of n .

52. $(n-1)!, n = 12$

53. $(n-1)!, n = 14$

54. Calculate the number of distinct Hamilton cycles in a complete graph with 13 vertices.

For the following exercises, use the figure shown to find the weight of each Hamilton cycle.

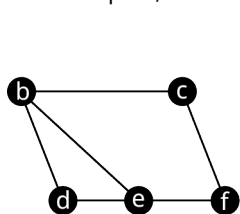


55. $q \rightarrow t \rightarrow w \rightarrow x \rightarrow u \rightarrow y \rightarrow v \rightarrow s \rightarrow r \rightarrow q$

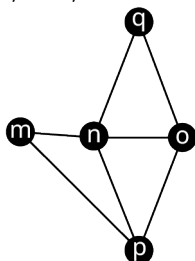
56. $w \rightarrow x \rightarrow y \rightarrow u \rightarrow v \rightarrow s \rightarrow r \rightarrow q \rightarrow t \rightarrow w$

Hamilton Paths

For the following exercises, use the figure shown to determine whether the sequence of vertices in the given graph is a Hamilton path, an Euler trail, both, or neither.



Graph A



Graph K

57. Graph A. $e \rightarrow b \rightarrow c \rightarrow f \rightarrow e \rightarrow b \rightarrow e$
 58. Graph A. $b \rightarrow c \rightarrow f \rightarrow e \rightarrow d \rightarrow b \rightarrow e$
 59. Graph K. $n \rightarrow q \rightarrow o \rightarrow p \rightarrow m$
 60. Graph K. $o \rightarrow q \rightarrow m \rightarrow n \rightarrow p$

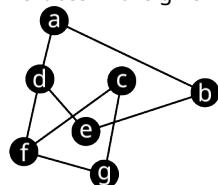
Recall the three common scenarios in which it is not possible to have a Hamilton between two vertices.

Scenario 1: If an edge ab is a bridge, then there is no Hamilton path between a pair of vertices that are on the same side of edge ab .

Scenario 2: If an edge ab is a bridge with at least three components on each side, then there is no Hamilton path beginning or ending at a or b .

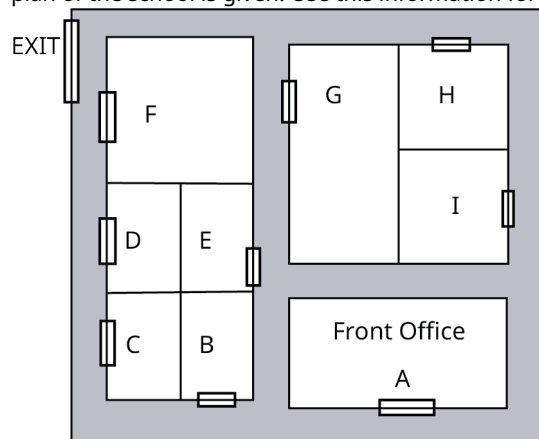
Scenario 3: If a graph is composed of two cycles joined only at a single vertex p , and v is any vertex that is *not* adjacent to p , then there are no Hamilton paths beginning or ending at p .

For the following exercises, identify the scenario that guarantees there is no Hamilton path between the given pair of vertices in the given graph.



61. Vertices a and e .
 62. Vertices d and f .

The principal of an elementary school plans to visit each classroom exactly once before leaving for the day. The floor plan of the school is given. Use this information for the following exercises.



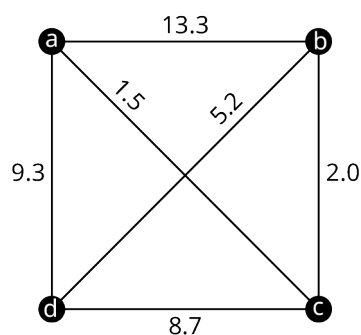
63. Draw a graph to represent the floor plan for the elementary school in which each vertex is a room and an edge between two vertices indicates that there is a path between the two rooms that does not pass the door to another room.
 64. Use your graph to find a route that the principal can take beginning at the Front Office in room A, visiting each room exactly once (without passing by another room), and ending at room F, which is next to the exit.
 65. Would the route you found be best to trace out a Hamilton path, an Euler trail, both, or neither in the graph you drew? Explain how you know.

Traveling Salesperson Problem

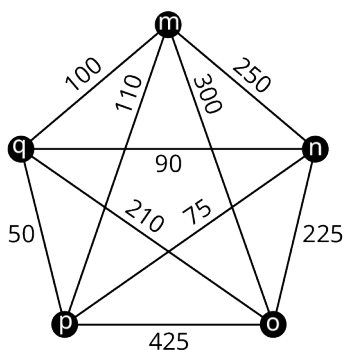
For the following exercises, determine whether the algorithm described is a greedy algorithm or a brute force algorithm.

66. A lumber distributor is loading pallets onto trucks with the intention of using the fewest trucks possible to send a shipment. The distributor loads the pallets with the greatest length that will fit on the truck first and continues loading until no more pallets will fit. Then the next truck is loaded using the same approach.
 67. A traveler wants to visit five cities by airplane. The traveler lists all the possible orders in which the cities can be visited then calculates the best airfare for each itinerary and selects the least expensive option.

For the following exercises, list all the distinct Hamilton cycles beginning at the given vertex in the given graph. Indicate which pairs of Hamilton cycles are reverses of each other.



Graph 17



Graph 18

68. Graph 17, vertex *a*69. Graph 18, vertex *m*

For the following exercises, find a Hamilton cycle of least weight for the given graph in Figure 12.362, beginning at the given vertex, and using the brute force method. What is the weight of the cycle?

70. Graph 18; vertex *m*71. Graph 17; vertex *a*

For the following exercises, find a Hamilton cycle of low weight for the given graph in Figure 12.362, beginning at the given vertex, and using the nearest neighbor method. What is the weight of the cycle?

72. Graph 18 vertex *m*73. Graph 17, vertex *a*

74. The products at a particular factory are manufactured in phases. The same equipment is utilized for each phase, but it must be formatted differently to accomplish different tasks. The transition time to convert between a format for one task and another task varies. The times are given in the table below.

Task	A	B	C	D	E	F	G
A	0	75	130	45	120	70	100
B	75	0	140	65	115	25	60
C	130	140	0	35	55	20	125
D	45	65	35	0	50	30	40
E	120	115	55	50	0	95	145
F	70	25	20	30	95	0	15
G	100	60	125	40	145	15	0

Use the table to draw a graph in which the vertices represent tasks A, C, D, F, and G, and the weights of the edges indicate the times in minutes to transition between pairs of tasks. Use the nearest neighbor method to find an order in which to complete tasks A, C, D, F, and G, which keeps the transition times down and ends with the same set up as it began so that the factory is ready start the next batch. Assume that there are no restrictions on which tasks can be completed in which order. Hint: The nearest neighbor algorithm may give a different result depending on which vertex is the starting vertex; so, you must check all possibilities.

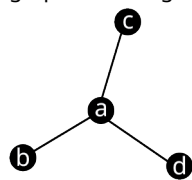
Trees

For the following exercises, draw a graph with the given characteristics.

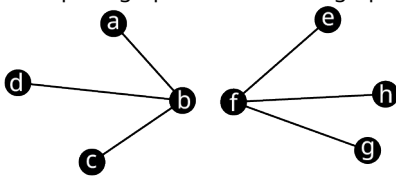
75. A tree with eight vertices, exactly two of degree three.

76. A connected graph with eight vertices, exactly two of degree 3, which is not a tree.

For the following exercises, identify each type of graph in the given figure: tree graph, star graph, starlike graph, line graph, lobster graph, caterpillar graph, and/or forest graph.



Graph 19

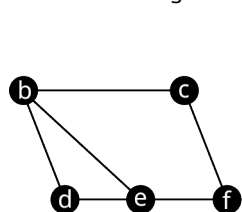


Graph 20

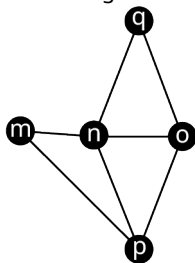
77. Graph 19

78. Graph 20

For the following exercises, use the figure shown to draw three possible spanning trees that fit the given description.



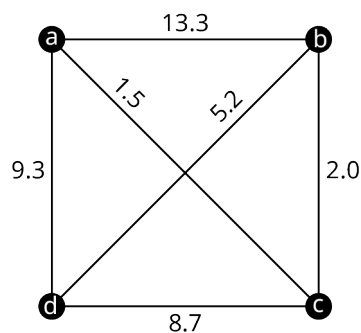
Graph A



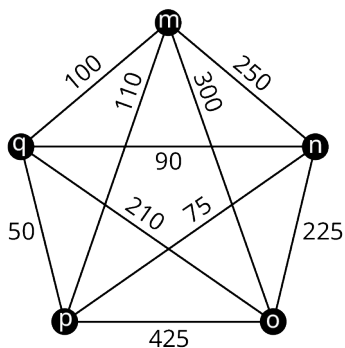
Graph K

79. Three spanning trees of Graph A, which include both edges be and de .80. Three spanning trees of Graph K, which include edges mn and oq , but do not include no .

For the following exercises, use Kruskal's Algorithm to find a minimum spanning tree of the given graph. Graph it and calculate its weight.



Graph 17



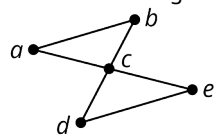
Graph 18

81. Graph 17

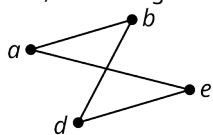
82. Graph 18

Chapter Test

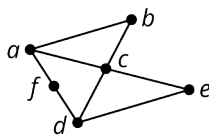
For the following Exercises, use the figure shown.



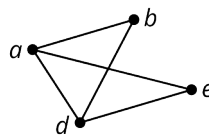
Graph S



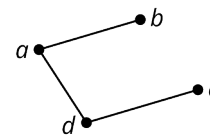
Graph T



Graph U



Graph V

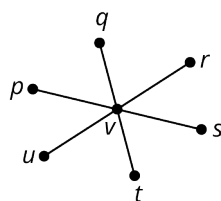


Graph W

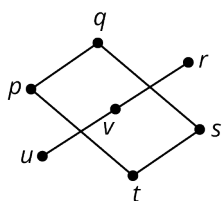
1. Name the edges in Graph T.

2. Identify the graph(s) with six edges.

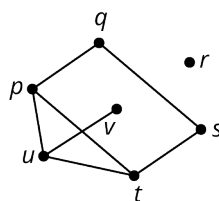
For the following exercises, use the figure shown.



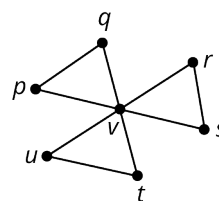
Graph A



Graph B

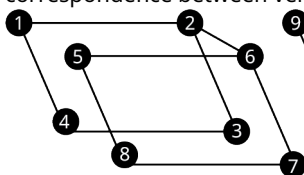


Graph C

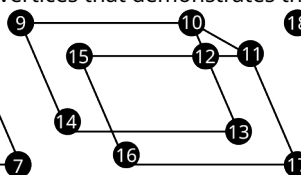


Graph D

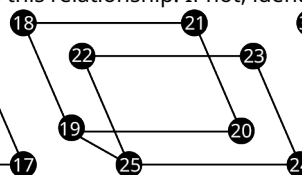
3. Identify any of the graphs that is a subgraph of Graph D.
4. Identify any graphs with no cyclic subgraphs of any size.
5. Consider Graph 4 and Graph 6 in the given figure. Determine if one graph is a subgraph of the other. If so, give a correspondence between vertices that demonstrates this relationship. If not, identify conflicting characteristics.



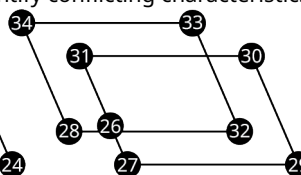
Graph 3



Graph 4

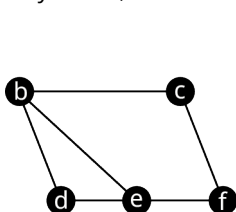


Graph 5

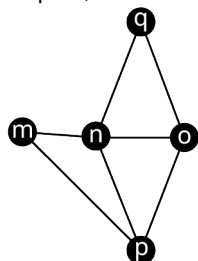


Graph 6

6. For the following exercise, use Graph A in the given figure. Consider each sequence of vertices. Determine if it is only a walk, both a walk and a path, both a walk and a trail, all three, or none of these.



Graph A

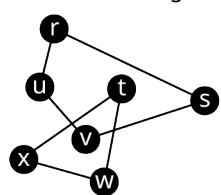


Graph K

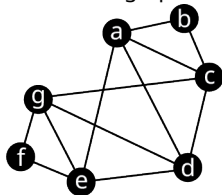
$d \rightarrow e \rightarrow b \rightarrow c \rightarrow f$

7. For the following exercise, use Graphs A and K in Figure 12.369. Determine the chromatic number of Graph K. Give a coloring that supports your conclusion.

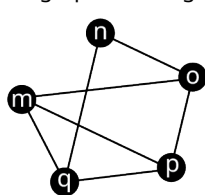
For the following exercise, use the graphs and multigraphs in the given figure.



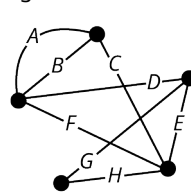
Graph 11



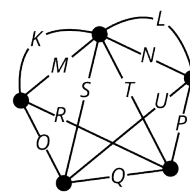
Graph 12



Graph 13

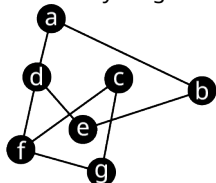


Multigraph 14

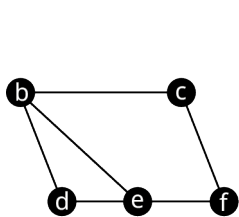


Multigraph 15

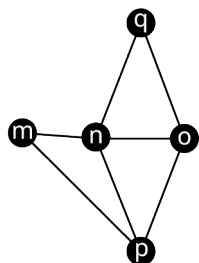
8. Identify any graphs and/or multigraphs that are not Eulerian. If there are none, state so.
9. List the set of vertices for each component in Graph 11.
10. Find an Euler circuit beginning and ending at vertex b in Graph 12 if one exists.
11. Use Fleury's algorithm to construct an Euler trail for the given graph beginning at vertex f of your choice.



12. Use the graphs shown to determine whether the sequence of vertices $d \rightarrow b \rightarrow c \rightarrow f \rightarrow e \rightarrow d$ is a Hamilton cycle, an Euler circuit, both, or neither.



Graph A

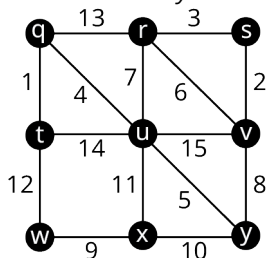


Graph K

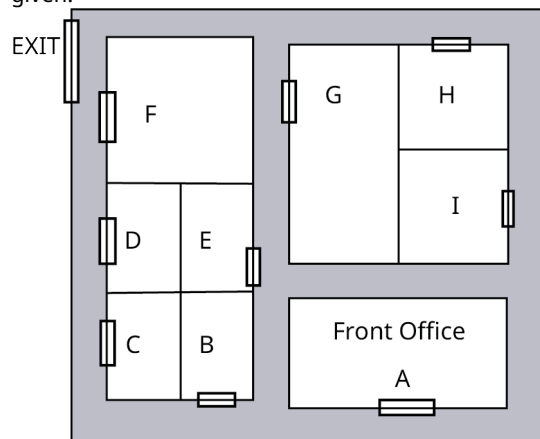
13. Calculate the number of distinct Hamilton cycles in a complete graph with 15 vertices.

14. Use the figure shown to find the weight of the given Hamilton cycle:

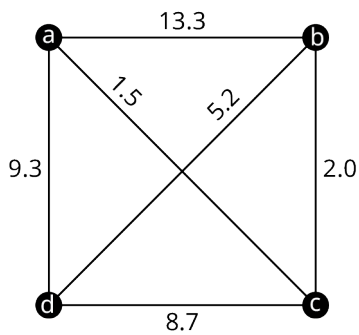
$t \rightarrow w \rightarrow x \rightarrow u \rightarrow y \rightarrow v \rightarrow s \rightarrow r \rightarrow q \rightarrow t$



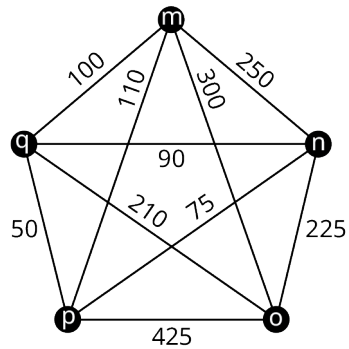
For the following exercises, use this information: For the Halloween celebration at an elementary school, the students from Classroom *I* will visit every classroom once and return to their own classroom. The floor plan of the school is given.



15. Draw a graph to represent the classrooms of the elementary school in which each vertex is a classroom and an edge between two vertices indicates that there is a path between the two rooms that does not pass the door to another classroom.
16. Use your graph to find a Hamilton circuit beginning and ending at *I*.
17. Explain what the Hamilton circuit you found represents for the students in Classroom *I*.
18. Find a Hamilton cycle of low weight for Graph 18, beginning at vertex *q*, and using the nearest neighborhood method. What is the weight of the cycle?



Graph 17



Graph 18

19. Use Kruskal's Algorithm to find a minimum spanning tree for the below graph. Graph the tree and give its weight.

